

Lecture 13: Characteristics vs Covariances

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Intro

- Last lecture gave us traded factors: market, SMB, HML, and momentum ([Fama and French 1993](#); [Carhart 1997](#))
- Today: what do those factors actually represent?
- Risk exposures?
- Or firm characteristics dressed up as factors?

- Part I: set up the characteristics-versus-covariances fight
- Part II: the Daniel-Titman challenge and the rebuttal ([Daniel and Titman 1997](#); [Davis, Fama, and French 2000](#))
- Part III: why empirical design can flip the answer ([Chordia, Goyal, and Shanken 2017](#); [Lewellen, Nagel, and Shanken 2010](#))
- Part IV: a modern reconciliation ([Kelly, Pruitt, and Su 2019](#))

Part I: What Is Being Priced?

- Fama and French (1993) showed that SMB and HML help price test portfolios
- But pricing success does not pin down economic interpretation
- A factor can work in the data for more than one reason

- Story 1: expected returns are driven by factor loadings
- Story 2: expected returns are driven by characteristics directly
- In the data, these two objects move together a lot

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i^\top f_t + \varepsilon_{i,t}$$

- If risk loadings are primitive, higher β_i should mean higher average returns
- Once we hold characteristics fixed, beta differences should still matter
- That is the Daniel-Titman test

Why Separation Is Hard

- High book-to-market firms often have different factor exposures
- Small firms often have different factor exposures
- Portfolio sorts change characteristics and covariances at the same time
- The empirical fight is really about identification

Part II: The Debate

- Core question: do returns follow characteristics or factor loadings? ([Daniel and Titman 1997](#))
- Hold size and book-to-market roughly fixed
- Then compare portfolios with different loadings on the Fama-French factors

Characteristic-Matched Portfolios

- Sort stocks on characteristics first
- Inside those buckets, form portfolios with different factor exposures
- This tries to break the usual link between characteristics and betas
- If FF risk is the whole story, return spreads should track loadings

- Add schematic: same size / BE-ME, different factor loadings

- Return differences line up more with characteristics than with loadings ([Daniel and Titman 1997](#))
- High book-to-market still earns more, even when factor exposures are similar
- That is a direct hit to the risk-based reading of SMB and HML

What Exactly Was Challenged?

- Not that factors forecast returns
- The challenged claim is deeper: are SMB and HML truly priced risks?
- Maybe factor models work because they proxy for characteristics

- Davis, Fama, and French (2000) push back on the characteristic story
- Their point: the value premium is robust over a longer sample
- They argue that covariance differences remain inside characteristic sorts
- The same evidence can still be consistent with factor pricing

Why the Rebuttal Matters

- Characteristics and covariances are empirically entangled
- If sorts do not isolate one from the other, interpretation stays open
- The debate is not settled by one successful portfolio experiment

Placeholder: Value Premium Comparison

- Add table or figure: characteristics story vs covariance story

- Move from sorted portfolios to individual stocks ([Chordia, Goyal, and Shanken 2017](#))
- Correct for errors-in-variables in estimated betas
- Betas regain explanatory power once the design changes
- Conclusion: the answer depends on the empirical protocol

What Changes When You Leave Portfolios?

- Portfolios reduce noise, but they also engineer factor structure
- Individual-stock tests are noisier, but closer to the raw cross-section
- This tradeoff becomes central for the whole literature

Questions?

Part III: Why Tests Break

- Many factor models appear to price the usual test assets ([Lewellen, Nagel, and Shanken 2010](#))
- Small alphas and high cross-sectional R^2 can be misleading
- False models can still look successful

Why Many False Models Look Good

- Test portfolios are highly correlated with each other
- Average returns live in a low-dimensional space
- Many different factor sets can span that space approximately
- Good fit is weaker evidence than it looks

$$\bar{R}_i = \hat{\lambda}^\top \hat{\beta}_i + \hat{\alpha}_i$$

- This second pass is fragile when $\hat{\beta}_i$ is noisy
- It is also fragile when test assets are constructed from the same characteristics
- The problem is power, not only specification

Placeholder: Low-Power Geometry

- Add figure: many models span the same small return space

- Apparent pricing success does not uniquely identify the right factors
- Empirical asset pricing can easily overproduce “good” models
- This is one reason the factor zoo arrives later

- Weak identification plus many candidate predictors is a dangerous combination
- Once researchers search widely, successful factors proliferate
- Next lecture: why the zoo got so large ([Harvey, Liu, and Zhu 2016](#); [McLean and Pontiff 2016](#))

Questions?

Part IV: Characteristics Are Covariances

- Kelly, Pruitt, and Su (2019) do not treat characteristics and covariances as rivals
- Characteristics forecast returns because they forecast latent factor loadings
- The debate becomes: how are betas measured?

A Unified Representation

$$\mathbb{E}_t[R_{i,t+1}] = \beta_{i,t}^\top \lambda_t, \quad \beta_{i,t} = b(z_{i,t})$$

- $z_{i,t}$ denotes firm characteristics
- Characteristics enter by shaping time-varying exposures
- They look “direct” only because the true factors are latent

- Daniel-Titman were right that characteristics predict returns
- Risk-based models were right that covariances matter
- Both can be true when characteristics proxy for loadings on latent factors

Placeholder: Characteristics to Covariances

- Add diagram: characteristics $\rightarrow$ latent betas $\rightarrow$ expected returns

- Daniel-Titman challenged the risk interpretation
- Davis-Fama-French and Chordia-Goyal-Shanken showed the answer depends on design
- Lewellen-Nagel-Shanken explained why many models seem to work
- Kelly-Pruitt-Su reconcile the debate
- Next step: the factor zoo

Readings and References i

- Carhart, Mark M. 1997. "On Persistence in Mutual Fund Performance." *The Journal of Finance* 52 (1): 57–82. <https://doi.org/10.1111/j.1540-6261.1997.tb03808.x>.
- Chordia, Tarun, Amit Goyal, and Jay A. Shanken. 2017. "Cross-Sectional Asset Pricing with Individual Stocks: Betas Versus Characteristics." <https://doi.org/10.2139/ssrn.2549578>.
- Daniel, Kent, and Sheridan Titman. 1997. "Evidence on the Characteristics of Cross Sectional Variation in Stock Returns." *The Journal of Finance* 52 (1): 1–33. <https://doi.org/10.1111/j.1540-6261.1997.tb03806.x>.
- Davis, James L., Eugene F. Fama, and Kenneth R. French. 2000. "Characteristics, Covariances, and Average Returns: 1929 to 1997." *The Journal of Finance* 55 (1): 389–406. <https://doi.org/10.1111/0022-1082.00209>.
- Fama, Eugene F., and Kenneth R. French. 1993. "Common Risk Factors in the Returns on Stocks and Bonds." *Journal of Financial Economics* 33 (1): 3–56. [https://doi.org/10.1016/0304-405X\(93\)90023-5](https://doi.org/10.1016/0304-405X(93)90023-5).
- Harvey, Campbell R., Yan Liu, and Heqing Zhu. 2016 "... And the Cross-Section of Expected Returns." *The Review of Financial Studies* 29 (1): 5–68. <https://doi.org/10.1093/rfs/hhv059>.

- Kelly, Bryan, Seth Pruitt, and Yinan Su. 2019. "Characteristics Are Covariances: A Unified Model of Risk and Return." *Journal of Financial Economics* 134 (3): 501–24. <https://doi.org/10.1016/j.jfineco.2019.05.001>.
- Lewellen, Jonathan, Stefan Nagel, and Jay Shanken. 2010. "A Skeptical Appraisal of Asset Pricing Tests." *Journal of Financial Economics* 96 (2): 175–94. <https://doi.org/10.1016/j.jfineco.2009.09.001>.
- McLean, R. David, and Jeffrey Pontiff. 2016. "Does Academic Research Destroy Stock Return Predictability?" *The Journal of Finance* 71 (1): 5–32. <https://doi.org/10.1111/jofi.12365>.